

ADA – Site Testing & Calibration

Configuring the ADA System On-Site at Nebico Oil Storage

RTD, Ultrasonic Level, VFD/Flow, and Oil-Node Output Calibration

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This is the working document for site testing and calibration of the ADA system at Nebico – the RTD temperature sensors, the ultrasonic level sensors, the VFD/flowmeter relationship, the tank geometry parameters (K_{100} and the mass-per-mm figure), and the oil-node output calibration constant. Tables are set out ready to fill in on site; the equations to derive from them are given alongside.

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1 Purpose and Scope

This document sets out how the ADA system is being tested and calibrated on site at Nebico, covering five distinct calibration jobs:

1. RTD temperature sensor calibration (all six sensors, re-done from scratch – Section 3).
2. Ultrasonic level sensor calibration against real tank fill levels (Section 4).
3. VFD-to-flowrate calibration for the oil pump, establishing the frequencies for LSM, HSM and FSM (Section 5).
4. Tank geometry parameters – the empty-tank value for $K100$, and the mass-per-mm conversion figure (Section 6).
5. The oil-node dispensed-volume calibration constant (Section 7).

Every table below is a template, not a result – it's meant to travel to site and come back filled in, with the fitted numbers transcribed into the PLC afterwards.

2 Why the Old RTD Data Is Being Scrapped

The RTD calibration in the previous Sensor Calibration Guide is being discarded, not revised. Two problems with how it was taken:

- **Non-steady-state sampling.** Readings were logged while the RTD value was still drifting toward its settled reading in the water bath, rather than waiting for it to level off first. A five-second average of a still-moving signal isn't a reading of the bath temperature – it's a reading of wherever the sensor happened to be on its way there.
- **Informal recording strategy.** There was no fixed protocol for how many readings to take, when to take them, or what counted as "settled" – points were logged more or less as convenient, which is how Sensor 6 ended up with a transcription error (72°C logged against a raw value of 722, read at the time as 122) that only surfaced later on inspection.

That data is not being patched or reused. Section 3 sets out a tighter protocol – with an explicit steady-state check – and all six RTD sensors are being re-measured against it from a blank sheet.

3 RTD Temperature Sensor Calibration

3.1 Method

All six RTD sensors are re-calibrated on the same two-point principle: an ice-water bath near 0°C and a hot-water bath, each checked against a calibrated mercury thermometer.

Steady-state check comes first. At each bath, watch the raw RTD value before logging anything. Only once it's approaching its settled value – no longer drifting, and no longer swinging through a wide margin scan to scan – does the actual logging sequence start. A reading taken while the value is still moving is not usable, however close it happens to land.

Once the value has settled, for *each* of the two baths (ice water, hot water):

1. Average the raw signal over five seconds (five one-second samples, the same cadence as the existing ADA sensor-averaging logic).

2. Take **five** such five-second averages in a row.
3. Average those five into a single point estimate for that bath.
4. Note the thermometer reading for the same bath alongside it.

Per sensor: 2 baths $\times$ 5 five-second averages = 10 raw samples feeding into 2 point-estimates, each point-estimate paired with one thermometer reading. Two points per sensor is enough to fit its own line directly – there’s no third anchor point in this round, since the steady-state check is what protects against a bad reading, not a third data point to catch one after the fact.

3.2 Per-Sensor Fit

For each RTD sensor, with the two paired points (x_1, T_1) (ice water) and (x_2, T_2) (hot water):

$$m_i = \frac{T_2 - T_1}{x_2 - x_1}, \quad c_i = T_1 - m_i x_1$$

giving each sensor $i = 1, \dots, 6$ its own distinct gain m_i and offset c_i .

3.3 Final Shared Equation

Once all six (m_i, c_i) pairs are in hand, they’re averaged across the six sensors to give one shared final equation:

$$\bar{m} = \frac{1}{6} \sum_{i=1}^6 m_i, \quad \bar{c} = \frac{1}{6} \sum_{i=1}^6 c_i, \quad T = \bar{m} x_{\text{raw}} + \bar{c}$$

This is the equation that goes into the ladder – one gain and one offset, applied across all six RTD channels – rounded so that T resolves to 0.1 °C, in keeping with the $\times 10$ fixed-point convention already used through the ADA register map.

After averaging, check each sensor’s individual error under $\bar{m}, \bar{c}$ against its own two points. If any sensor sits noticeably further off the shared line than the rest, that’s worth a repeat pair of baths for that sensor specifically, before accepting the shared equation as final – the same way the old data flagged Sensors 1 and 6 as outliers worth a second look.

3.4 Data Template

Table 1: RTD calibration – template, one row set per sensor.

Sensor	Bath	Thermometer (°C)	Raw avg. (5×5s)	m_i	c_i
1	Ice water				
1	Hot water				
2	Ice water				
2	Hot water				
3	Ice water				
3	Hot water				
4	Ice water				
4	Hot water				
5	Ice water				
5	Hot water				
6	Ice water				
6	Hot water				

$$\bar{m} = \underline{\hspace{2cm}} \quad \bar{c} = \underline{\hspace{2cm}}$$

4 Ultrasonic Level Sensor Calibration

4.1 Method

Unlike the earlier round – which sighted the level sensors sideways at a control panel cover and a chair, then had to convert a distance-to-target figure into a tank level via a separately measured offset h – this round calibrates each sensor directly against the tank it will actually be reading, at three real fill states:

- **Empty tank**
- **Half-filled tank**
- **Fully filled tank**

At each fill state, the true level is the known physical quantity (empty = 0, full = the tank’s full height, half = half that height, or whatever level is independently confirmed by dip-stick or sight glass), and the corresponding raw sensor value is logged the same way as the RTD readings – averaged over five seconds, taken as multiple consecutive averages once the reading has settled. The reading obtained is supposed to be inverse of the original, The full tank will give low value and empty tank will give high value, but that calculation is to be handled by the site testing engineer. At best, if no breakthrough can occur, we can traverse back to the original method where the full height is measured first then is subtracted from the obtained value to get the level fill value.

This is expected to be the steadier, more reproducible of the two sensor types to calibrate – a real liquid surface in a tank is a far more consistent target than a panel cover and a chair positioned by hand, so three real fill points should sit close to a straight line without the scatter the previous round saw.

Because the calibration is run directly against tank level rather than against sensor-to-target distance, the fitted line gives the level *directly* – there’s no separate geometric offset (h) to subtract afterwards, unlike the previous method.

4.2 Fit

$$L = m \cdot x_{\text{raw}} + c$$

fitted from the three (raw, level) pairs per sensor – least-squares across the three points, the same form used elsewhere in the ADA ladder.

4.3 Data Template

Table 2: Level sensor calibration – template, one row set per sensor (repeat for all level sensors in use).

Fill state	True level (mm)	Raw avg. (5s, settled)	Notes
Empty			
Half-filled			
Fully filled			
		$m =$ _____	$c =$ _____

5 VFD-to-Flowrate Calibration

5.1 Method

The oil pump’s VFD is calibrated against measured discharge velocity using a bucket and a timer – no flowmeter reference is needed, since the bucket-and-timer measurement *is* the reference.

1. Pick a bucket size that suits the discharge rate at hand – big enough that the fill time is easy to time accurately, small enough that it fills in a reasonable window.
2. At a given VFD frequency, discharge into the bucket and time the fill. Flowrate follows from volume over time; velocity follows from flowrate over the pipe’s cross-sectional area (pipe diameter 1.5 in):

$$Q = \frac{V_{\text{bucket}}}{t}, \quad v = \frac{Q}{A_{\text{pipe}}}, \quad A_{\text{pipe}} = \frac{\pi}{4} (1.5 \text{ in})^2$$

3. Adjust the VFD frequency and repeat until the measured velocity lands on each of the three target speeds below.

Three target speeds to find frequencies for:

Mode	Target velocity	Rounded to
LSM (Low Speed Mode)	0.533 m/s	0.5 m/s
HSM (High Speed Mode)	1.067 m/s	1.0 m/s
FSM (Full Speed Mode)	1.2 m/s	1.2 m/s

For each mode, the VFD frequency that produces the rounded target velocity (within whatever tolerance the bucket-and-timer method can resolve) is the frequency written into the PLC/VFD parameters for that mode. The Site engineer should connect the PLC to VFD in a way that when the PLC outputs through Y2, Y3 and Y4; the VFD should produce the frequency that outputs 0.5 m/s, 1.0 m/s and 1.2 m/s. (As specified in Electrical Schematic produced earlier).

5.2 Data Template

Table 3: VFD frequency vs. measured velocity – template.

Trial	VFD freq. (Hz)	Bucket vol.	Fill time (s)	Velocity (m/s)	Target reached?
					LSM (0.5 m/s)
					HSM (1.0 m/s)
					FSM (1.2 m/s)
LSM freq. = _____ Hz HSM freq. = _____ Hz FSM freq. = _____ Hz					

6 Tank Geometry Parameters

6.1 Empty-Tank Value for K100

Once the tank height has been measured on site, $K100$ – the constant currently sitting in the ladder as a placeholder – is replaced with the sensor value actually read when the tank is confirmed empty, taken from the same settled, averaged reading used throughout this document, not a single instantaneous sample.

Measured tank height: _____ mm Empty-tank raw/level value ($K100$): _____

6.2 Mass per mm

From the tank's diameter, measured on site, the mass represented by each millimetre of tank height is calculated directly rather than assumed:

$$V_{\text{mm}} = \frac{\pi}{4} D_{\text{tank}}^2 \times 1 \text{ mm}, \quad \frac{\Delta m}{\Delta h} = V_{\text{mm}} \times \rho_{\text{oil}}$$

where D_{tank} is the on-site tank diameter and ρ_{oil} is the palm oil density at the system's operating temperature. This figure feeds directly into the program's mass-conversion parameter.

Tank diameter D_{tank} : _____ mm Oil density ρ_{oil} at operating temp.: _____ kg/L

Mass per mm ($\Delta m / \Delta h$): _____ kg/mm

7 Oil-Node Output Calibration Constant

7.1 Method

With Tank T5 holding a reasonable amount of oil, the tester commands the oil node to dispense 5 L and measures what’s actually delivered using a bucket.

$$K_{\text{new}} = \frac{5}{V_{\text{measured}}}$$

where V_{measured} is the actual dispensed volume, in litres, read off the bucket for a commanded 5 L.

This is repeated – adjusting the constant and re-testing – until the error settles as low as practically achievable, with 0.5 L as the maximum acceptable error at any tested volume. The check isn’t limited to the 5 L point: other commanded volumes (e.g. 10 L, 15 L, and further steps as convenient) should also be run against the updated constant, to confirm it holds up across the range the oil node is actually used at, not just at the one calibration point.

7.2 Data Template

Table 4: Oil-node dispensed-volume calibration – template.

Trial	Commanded vol. (L)	Measured vol. (L)	Error (L)	K_{new}
1	5			
2	5			
3	5			
4	10			
5	15			

Keep repeating trials at 5 L until the error is at or below the 0.5 L ceiling before moving on to check the other volumes – a constant that passes at 5 L but drifts wider at 10 L or 15 L is worth another look at whether the relationship is really linear across the node’s working range, rather than accepting the 5 L point alone as sufficient.

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